

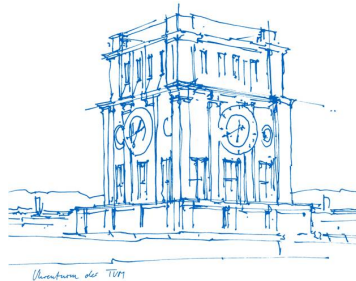
HeatPath–GNN

Learning where to place conductive particles

Thermal transport through
particle contact networks

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Outline

Motivation and research question

Packing and thermal model

Allocation methods and teacher search

Graph learning and prediction

Results and conclusions

The story moves from the physical allocation problem to fast prediction on an unseen packing.

Section 1 of 5

Motivation and research question

A fixed material budget turns heat transfer into an allocation problem

Motivation: conductive material is a limited resource

Application examples

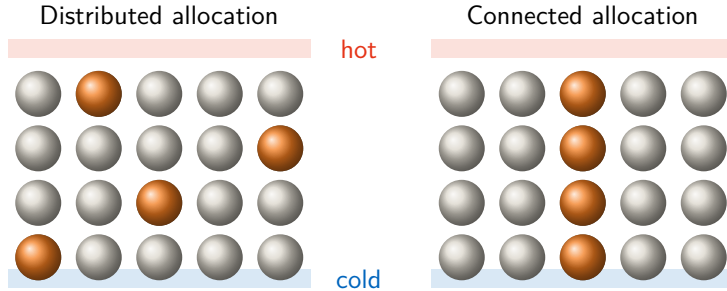
- **Catalytic reactors:** conductive inert pellets spread heat but replace active catalyst.
- **Adsorption beds:** graphite improves heat removal but replaces adsorbent.
- **Thermal-storage beds:** conductive filler speeds charging but adds cost or replaces storage material.

Common design constraint

- The conductive fraction is limited.
- Packing geometry and size distribution remain fixed.
- Particle placement becomes the design variable.

Where should the limited conductive particles be placed to maximize heat conduction?

The allocation problem



Orange: high conductivity. Grey: low conductivity. Schematic only.

Geometry and material amount stay fixed. Only particle labels change.

Section 2 of 5

Packing and thermal model

Particle contacts define the available heat-transfer routes

Thirty packings with the same nominal porosity

Quantity	Value
Particles	500
Diameters [mm]	1.5 / 2.0 / 2.5
Counts	100 / 300 / 100
Nominal porosity	0.38
Independent seeds	30

- Periodic boundaries in x and y
- Two fixed thermal walls in z
- No gravity during generation
- Grow small spheres, then relax
- Accept final translational $KE \leq 10^{-12}$ J

DEM: Hertz–Mindlin, $E = 50$ MPa, $\nu = 0.25$, restitution 0.30, friction 0.50.

Contact conductance links temperature difference to heat rate

Conductance: units W/K

$$Q_{ij} = G_{ij}(T_i - T_j)$$

G_{ij} : between particles i and j .

G_{iw} : between particle i and wall w .

Q_{ij} : heat rate [W].

Material conductivity k has units W/(m K).

Steady balance at particle i

$$\sum_j G_{ij}(T_i - T_j) + \sum_w G_{iw}(T_i - T_w) = 0$$

Whole-bed conductivity

$$k_{\text{eff}} = \frac{QH}{A(T_{\text{hot}} - T_{\text{cold}})}$$

Q : wall heat rate, H : height, A : wall area.

A larger conductance transfers more heat for the same temperature difference.

Thermal conductance follows contact size and material

Particle–particle contact

$$a_{ij} = \sqrt{R_{ij}^* \delta_{ij}}, \quad R_{ij}^* = \frac{R_i R_j}{R_i + R_j}$$

$$G_{ij} = \frac{4a_{ij}k_i k_j}{k_i + k_j}$$

δ_{ij} : geometric overlap.

a_{ij} : Hertz contact radius.

Ideal circular constriction model. Its small-contact assumption needs physical assessment.

Material assignment

$$k_i \in \{1, 10\} \text{ W m}^{-1}\text{K}^{-1}$$

Isothermal wall

$$G_{iw} = 4a_{iw}k_i$$

Two spreading resistances in series give the particle-contact formula.

Section 3 of 5

Allocation methods and teacher search

Boundary conditions, baseline solve, allocation methods and swap optimization

Thermal experiment and fixed material budget

Boundary conditions

- Bottom wall: 301 K
- Top wall: 300 K
- Periodic in x and y
- Steady contact conduction

Diameter [mm]	1.5	2.0	2.5
Total particles	100	300	100
High- k particles	10	30	10

$$\max_{\mathbf{s}} k_{\text{eff}}(\mathbf{s}), \quad s_i \in \{0, 1\}, \quad \sum_{i \in \mathcal{B}_b} s_i = q_b$$

Here $s_i = 1$ assigns high conductivity to particle i . The set $\mathcal{B}_b$ contains one size class, and q_b fixes its number of high- k particles. We change only the assignment $\mathbf{s}$ and seek the largest k_{eff} .

Fifty of 500 particles are conductive, with exactly 10% in each size class.

All-low thermal solve before material allocation

Step 1: assign the low conductivity to every particle

$$k_i = k_{\text{low}} = 1 \text{ W m}^{-1} \text{K}^{-1} \quad \text{for every particle}$$

Baseline temperature $T_i^{(0)}$

Solve the contact network between the 301 K bottom wall and the 300 K top wall. Each particle receives a baseline temperature.

Baseline heat-throughput score

$$s_i^{(0)} = \sum_{j \in \mathcal{N}(i)} \left| G_{ij}^{(0)} (T_i^{(0)} - T_j^{(0)}) \right|$$

Sum the absolute heat rates at particle–particle contacts [W].

The all-low solve supplies thermal features before any high- k particles are selected.

Reference methods select which 50 particles become conductive

Method	Selection rule within each size class
Random allocation	Draw 10 / 30 / 10 particles at random. Repeat 100 times and average the resulting bed conductivities.
Degree ranking	Count each particle's contacting neighbours. Select the 10 / 30 / 10 particles with the highest counts.
Heat-throughput ranking	First solve the all-low-conductivity bed. Select the 10 / 30 / 10 particles with the largest baseline heat-throughput scores.

Each allocation gives a list of high-conductivity particle IDs, then a thermal solve gives k_{eff} .

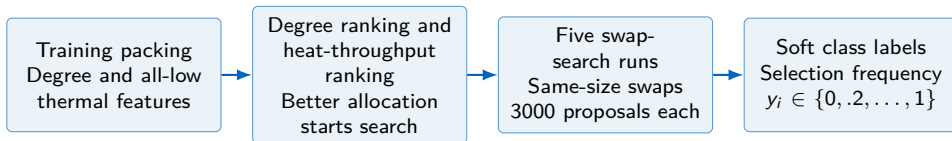
Swap search starts from the better reference allocation

1. Evaluate the allocations from degree ranking and heat-throughput ranking. Start from whichever gives the larger k_{eff} .
2. Randomly choose a size class, then one high- k particle and one low- k particle from that class.
3. Exchange their material labels and solve the thermal network.
4. Keep the swap if k_{eff} increases. Otherwise keep the previous allocation.

Five runs use the same starting allocation and different random swap sequences.

Each run proposes 3000 swaps. Retain the best final allocation found.

Teacher allocations for model training



Search output

Each particle receives a target equal to the fraction of final search runs that selected it.

Scientific references

Random, degree and heat-throughput allocations remain independent benchmarks for the learned models.

The searched allocations provide training labels. They are not required when the trained GNN is applied.

Degree and heat throughput have three distinct roles

Role	How the quantities are used	Output
1. Initialize search	Degree and heat-throughput rankings each select particles.	Better allocation starts swap search.
2. Supply features	GNN and MLP use both values during training and prediction.	Learned particle scores.
3. Compare methods	Evaluate both rankings on the held-out packing.	Reference k_{eff} values for comparison with GNN.

Feature: a value for each particle. Ranking method: a rule for selecting particles.

Section 4 of 5

Graph learning and prediction

The GNN learns a node-level classification score for material allocation

The GNN uses geometry and the all-low thermal solution

Eight particle features

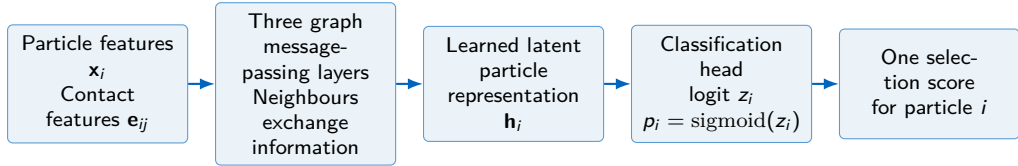
- Diameter, height and contact degree
- Hot-wall, cold-wall and anchored flags
- Normalized all-low temperature
- Normalized all-low heat-throughput score

Six contact features

- Contact radius
- Relative displacement in x, y, z
- Absolute vertical alignment
- Normalized absolute all-low contact heat rate

Three message-passing layers, width 64. MLP uses the same particle features. GNN additionally uses neighbour messages and contact features.

The GNN performs particle classification



Training question

How closely does this particle resemble particles retained by the teacher search?

Model output

A continuous priority score for every particle, not the bed conductivity.

This is node-level soft classification followed by constrained ranking.

Soft classification targets come from the search

Target for particle i

$$y_i = \frac{\text{final search runs selecting } i}{5}$$

Selection in four of five runs gives $y_i = 0.8$.

The target expresses search agreement, not a proven probability of global optimality.

Classification and ranking loss

$$L = L_{\text{ranking}} + 0.2L_{\text{BCE}}$$

BCE compares the sigmoid score p_i with the soft label y_i .

The ranking term favours particles with larger y_i within the same size class.

The network learns relative particle priorities. It does not regress k_{eff} .

Ranking within size classes enforces the budget

The GNN scores all 500 particles

- One continuous score per particle
- Three trained repeats give three rankings
- Average the ranks within each size class

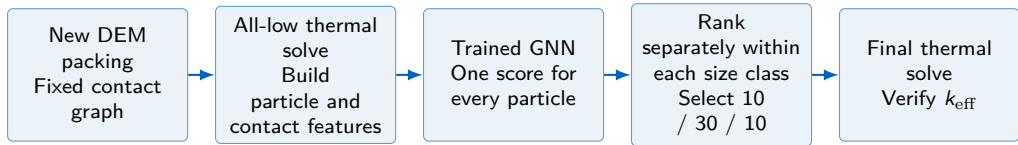
Selection remains stratified

- Top 10 among the 1.5 mm particles
- Top 30 among the 2.0 mm particles
- Top 10 among the 2.5 mm particles

$$\mathcal{S}_b = \text{Top}_{q_b} \{s_i : i \in \mathcal{B}_b\}, \quad \mathcal{S} = \bigcup_b \mathcal{S}_b$$

There is no global top-50 ranking across different particle sizes.

Applying the trained GNN to a new packing



Required for prediction

One reference solve, feature construction, GNN scoring and quota selection.

Required only for evaluation

Random, degree, heat-throughput and full swap-search comparisons.

The trained GNN replaces the repeated swap search on the unseen packing.

Each test packing is withheld from model fitting

28 training packings
weights and scalars

1 validation packing
early stopping

1 test packing
reported allocation

- Repeat for all 30 test packings
- Up to 400 epochs and patience 60
- Compare methods on the same held-out geometries

This tests transfer between realizations of the specified packing.

Section 5 of 5

Results and conclusions

Graph information improves allocation with far fewer thermal solves

Gain and recovery answer different questions

$$\text{Gain}_m = 100 \frac{k_m - k_{\text{random}}}{k_{\text{random}}}$$

$$\text{Recovery}_m = 100 \frac{k_m - k_{\text{random}}}{k_{\text{search}} - k_{\text{random}}}$$

Gain

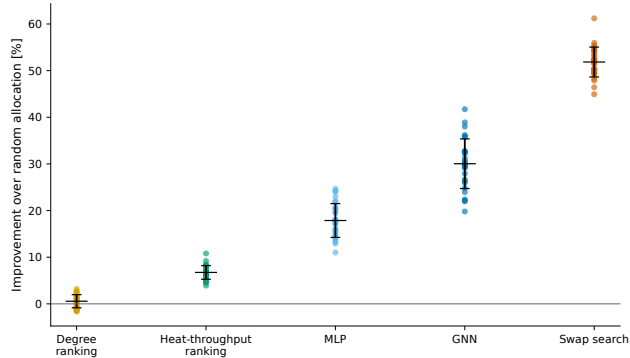
How much better is this allocation than random?

Recovery

How much of the search improvement does the model obtain?

Compute both per packing, then average. Error bars describe sample SD across packings.

GNN improves conductivity by 30% over random

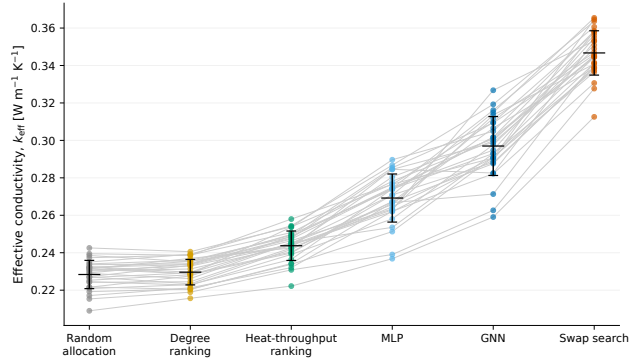


Mean improvement

Degree ranking	0.6%
Heat-throughput ranking	6.7%
MLP	17.9%
GNN	30.0%
Swap search	51.8%

Same geometry and material budget for every comparison.

Absolute conductivity remains below the search reference

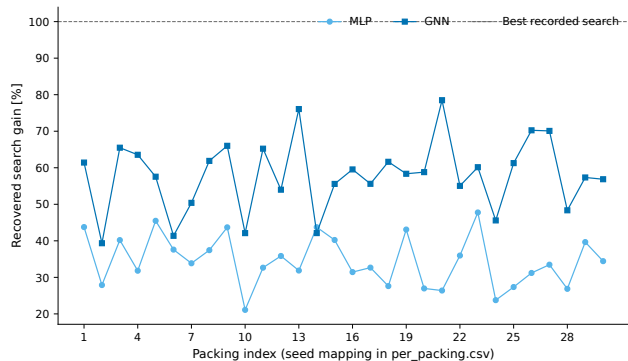


Mean k_{eff} [$\text{W}/(\text{m K})$]

Random allocation	0.2284
MLP	0.2692
GNN	0.2970
Swap search	0.3467

Grey lines connect results on the same packing.

GNN beats MLP on 29 of 30 held-out packings



Recovered search improvement

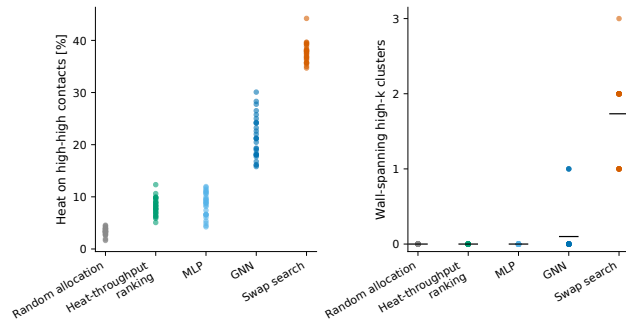
GNN: **58.0%**

SD: 9.8 percentage points

MLP: 34.5% mean

Improvement is consistent, but performance varies between packings.

Contact-network diagnostics help interpret allocation

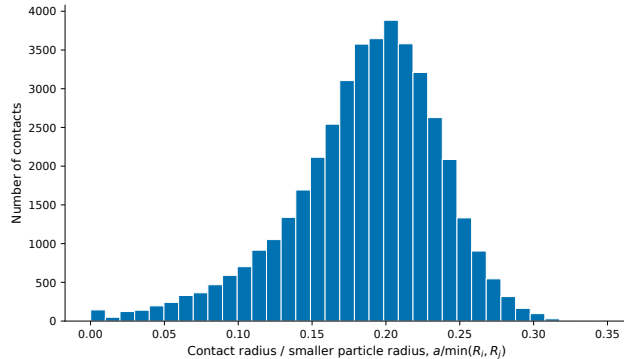


Two diagnostics

- Fraction of absolute contact heat carried by high–high contacts
- Number of high-conductivity clusters spanning both walls

These describe the selected network. They do not prove causality.

Numerical checks pass, but contact sizes need scrutiny



Recorded checks

- Maximum final KE:
 $8.4 \times 10^{-13} \text{ J}$
- Maximum energy imbalance:
 9.01×10^{-12}

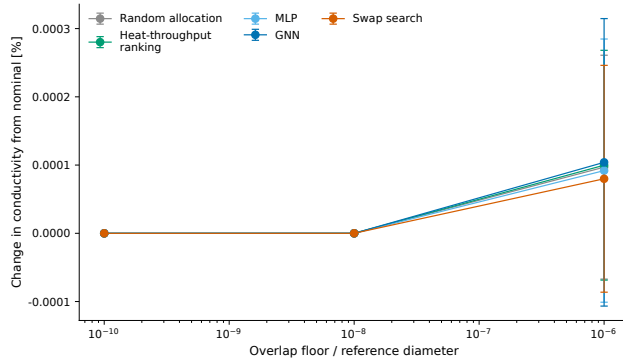
Contact radius / smaller radius

Median: 0.192

95th percentile: 0.258

Maximum: 0.347

The numerical overlap floor has little influence



Fixed allocations

- Vary the numerical overlap floor
- Largest absolute conductivity change: 0.000923%
- Sensitivity is negligible over the tested range

This does not test stiffness, loading or alternative contact laws.

Fast scoring is one part of the complete prediction procedure

What the reported times cover

- Score all particles with trained models
- Mean GNN ensemble scoring: 8.63 ms
- Mean MLP ensemble scoring: 0.295 ms

Additional work for a new packing

- Solve the all-low-conductivity baseline
- Build particle and contact features
- Combine ranks and enforce quotas
- Solve the selected thermal network

Search uses about 14,964 thermal solves per packing. Training is an additional upfront cost.

Scoring time alone is insufficient to claim an end-to-end speedup.

Conclusions and next research steps

What the campaign establishes

- Material placement strongly affects modelled conductivity
- GNN gains 30% over random allocation
- Neighbour information improves on the MLP
- 58% of best-found search gain is recovered

What remains to establish

- Physical sensitivity of contact areas
- Transfer to other packing specifications
- GNN-initialized search versus random starts
- Complete runtime and manufacturing feasibility

A learned particle ranking captures useful thermal-network structure.